

ROADNETRA AI · MODEL SPECIFICATION

Model B: Severe Accident Detection (Google Colab Guide)

Event: IEEE Hackathon 2026
Track: Gen AIML
Problem: 03.1 Vision in the Wild
Team: ESPADA

Mission Context: This document defines the exact architecture, data augmentations, and training hyperparameters for **Model B (Severe Accident Detection)** being trained on Google Colab. Following these specifications guarantees 100% plug-and-play compatibility when dropping `accident_best.pt` into our local RoadNetra Dual-Stream Video Dispatcher.

1. Project Architecture Overview (Dual-Stream Decoupled Engine)

To satisfy the 'Vision in the Wild' mandate while preventing GPU latency bottlenecks and dataset label interference, RoadNetra AI decouples road monitoring into two specialized vision streams running over shared CCTV footage:

Stream Attribute	Model A: Road Infrastructure (Local)	Model B: Severe Accidents (Colab)
Target Hazards	Potholes, surface cracks, broken road furniture	Vehicle rollover, high-speed impact crash, vehicle fires
Operational SLA	Municipal Maintenance (Hourly / Daily log)	Emergency Life-Safety Dispatch (< 2 seconds)
Inference Frequency	Sampled 1 – 2 FPS (Every 15th–30th frame)	Full 30 FPS (Evaluated on every incoming frame)
Model Architecture	Ultralytics YOLOv8s (Fine spatial detail)	Ultralytics YOLOv8s or YOLOv8n (Low-latency real-time)
Authority Routing	Municipal PWD / NHAI Highway Maintenance	Emergency 112 / Police / Hospital Ambulance Services

2. Model B Technical Specifications & Synchronization Contract

To ensure seamless loading via YOLO('models/accident_best.pt') on the local station, the Colab model must satisfy these exact parameters:

Parameter	Required Value	Technical Rationale
Model Architecture	Ultralytics YOLOv8s (or YOLOv8n)	Identical architecture family for PyTorch & MPS edge runtime.
Framework Version	ultralytics >= 8.1.0, torch >= 2.1	Guarantees serialization compatibility without missing custom op errors.
Input Image Dimension	640 x 640 (RGB 3-channel)	Matches video pre-processing letterboxing buffer.
Class Label & ID	Class 0: severe_accident	Zero collision with Model A (where Class 0=Pothole, 1=Road Damage).
Export File Name	accident_best.pt	Automated loading target in RoadNetra's Dual-Stream Engine.
Latency Target	≤ 25 ms (GPU) / ≤ 75 ms (CPU)	Enables sustained ≥30 FPS CCTV feed processing without dropped frames.

3. 'Vision in the Wild' Operational Robustness Requirements

Hackathon Problem 03.1 explicitly mandates resilience under real, messy operating environments. The accident model must be trained to overcome six core operational hazards:

A. Severe Illumination Variations & Night Glare

Field Challenge: CCTV cameras encounter extreme lighting shifts: pitch-black rural highways, blinding oncoming vehicle high-beams, and deep flyover shadows.

Strategy: Applied ML Mitigation: Configure brightness jitter (hsv_v=0.4) and saturation jitter (hsv_s=0.7). Include night crash images with emergency strobe light flares.

B. Camera Vibration, Sway & Motion Blur

Field Challenge: Overhead poles sway in heavy winds and vibrate from heavy commercial traffic; moving patrol vehicle dashcams experience high road vibration.

Strategy: Applied ML Mitigation: Inject synthetic Gaussian and motion blur (blur=0.02). Ensure training data includes dynamic frames with vehicle motion trails.

C. Heavy Occlusion & Cluttered Environments

Field Challenge: Severe collisions are frequently obscured by stopped traffic queues, concrete jersey barriers, guardrails, roadside vegetation, or billowing smoke.

Strategy: Applied ML Mitigation: Enable full mosaic=1.0 and mixup=0.15. Forces YOLO to extract accident cues from partial bumper, bent axle, or flipped chassis fragments.

D. Weather Noise (Monsoon Rain, Smog & Dirty Domes)

Field Challenge: Rain droplets streaking CCTV lens glass, road spray, thick fog, and environmental grime smudge vehicle silhouettes and lower contrast.

Strategy: Applied ML Mitigation: Apply contrast shifts, simulated water glare, and haze augmentations during dataset synthesis to prevent false negatives during monsoon downpours.

E. Extreme Elevation Angles & Perspective Distortion

Field Challenge: Highway high-mast surveillance cameras and junction fisheye domes capture vehicles at steep oblique angles with distorted proportions.

Strategy: Applied ML Mitigation: Activate perspective warp (perspective=0.0005), random rotation (degrees=10.0), and scaling (scale=0.5) in YOLO config.

F. False Alarm Suppression (Traffic Jams vs Crashes)

Field Challenge: Dense bumper-to-bumper peak hour traffic jams with stationary vehicles must NEVER trigger false positive accident alerts.

Strategy: Applied ML Mitigation: Supply a dedicated Negative Control Set (200+ images of congested rush-hour traffic containing intact vehicles labeled with 0 bounding boxes).

4. Recommended Open Datasets for Model B

Dataset Source	Target Hazard Types	Access Recommendation
CADP (Car Accident Detection)	1,416 CCTV video clips with annotated collision bounding boxes	Primary CCTV reference dataset
Roboflow Universe (Accidents)	Overturned cars, multi-vehicle pileups, vehicle fires	Search: car-accident-detection
Negative Control Set	150–200 frames of intact dense traffic jams & empty highways	Empty text annotation files (.txt with 0 boxes)

5. Google Colab Implementation (Copy-Paste Script)

Execute these snippets inside Google Colab (with T4 GPU runtime enabled) to fine-tune and export the model:

Step 5.1: Dataset Configuration (data.yaml)

```
path: /content/accident_dataset # Dataset root folder in Colab
train: images/train # Training images directory
val: images/val # Validation images directory

names:
0: severe_accident # Standardized single-class label
```

Step 5.2: Colab Training Cell with 'Wild' Augmentations

```
# 1. Install Ultralytics
!pip install -q ultralytics

from ultralytics import YOLO
import torch
print(f'Training on GPU: {torch.cuda.get_device_name(0)}')
```

2. Load Pretrained Checkpoint (Transfer Learning)

```
model = YOLO('yolov8s.pt') # or 'yolo11s.pt'
```

3. Execute Training with Vision-in-the-Wild Augmentations

```
results = model.train(
    data='data.yaml',
    epochs=40, # 30-50 epochs is ideal for fine-tuning
    imgsz=640, # 640x640 standard resolution
    batch=16,
    patience=10, # Early stopping if validation loss plateaus
    device=0,
    # --- 'Vision in the Wild' Augmentation Hyperparameters ---
    hsv_h=0.015, # Hue adjustments
    hsv_s=0.7, # Saturation jitter (smog & heavy rain)
    hsv_v=0.4, # Brightness jitter (night CCTV, tunnels, bright sun)
    degrees=10.0, # Pole sway & camera tilt rotation
    translate=0.1, # Frame boundary shifts
    scale=0.5, # Distance scale variations (close vs distant crashes)
    perspective=0.0005, # Oblique highway perspective lines
    mosaic=1.0, # Occlusion handling & multi-vehicle collisions
    mixup=0.15, # Blends background noise into scene
    # -----
    project='roadnetra_accident',
    name='yolov8s_wild_v1',
    exist_ok=True
)
```

4. Validate Metrics & Benchmark Inference Latency

```
metrics = model.val()
print(f'Validation mAP50: {metrics.box.map50:.4f}')
print(f'Validation mAP50-95: {metrics.box.map:.4f}')
```

5. Download Weights for Local RoadNetra Engine

```
from google.colab import files
files.download('roadnetra_accident/yolov8s_wild_v1/weights/best.pt')
```

6. Local Handoff & Synchronization Checklist

1. Rename the downloaded Colab file from best.pt to **accident_best.pt**.
2. Place accident_best.pt into the local project directory under **models/**.
3. RoadNetra's Dual-Stream Engine automatically binds Model B to the **30 FPS real-time emergency queue**.
4. Our **Temporal Smoothing Filter** validates detections across ≥ 3 consecutive frames (conf ≥ 0.55), eliminating shadow/glare false alerts and automatically triggering Police/Emergency dispatch cards.